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Department of Computer Science & Engineering

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**Minor Project Presentation on
AI driven Resume Screening System**

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Objectives

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1. **Automate Resume Screening:** Reduce manual effort in filtering candidates by automating the process using AI.
 2. **Enhance Efficiency:** Speed up the resume screening process, allowing HR teams to focus on higher-value tasks.
 3. **Improve Accuracy:** Use data-driven algorithms to make more accurate decisions about candidate qualifications.
 4. **Cost Reduction:** Decrease the operational costs associated with hiring, especially in large-scale recruitment drives.
 5. **Enhanced Candidate Experience:** Provide faster feedback to applicants by streamlining the screening process.
 6. **Data Utilization:** Leverage past hiring data to improve the model's accuracy over time.
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Existing System (Literature Survey)

Manual Resume Screening: Currently, recruiters manually screen resumes, which is time-consuming and prone to human errors.

Existing Automation Tools:

Applicant Tracking Systems (ATS): These tools use keyword matching to filter resumes but often miss important context and nuances eg- Enhancv, Jobscan

Tools like **HireVue** and **LinkedIn Talent Insights** use AI, but many still rely on predefined rules instead of sophisticated learning algorithms.

AI-based Systems: Some companies have begun implementing AI for resume screening, but these systems are often complex, expensive, or not customizable to small and medium-sized enterprises.

Challenges with Current Systems:

Lack of adaptability to different job roles.

Limited NLP capabilities to handle unstructured resume formats.

Potential bias in automated systems due to biased training data.

Existing System vs Our Idea

Existing Applications	Our Ideation
Limited to detecting predefined elements like grammatical errors, syntax issues, and keyword-based requirements.	Produces result by analyzing holistically and ranks the resumes according to scores
Provides standard responses without tailored evaluations for specific job roles.	Analyzes entire resume with ML algorithms.
Customized resume detection systems are expensive and unaffordable for many organizations.	We are offering the services free of cost or at affordable costs.
Don't have features to verify the knowledge of candidate for standard notion.	Have the skills verification feature.

Methodology

Data Collection

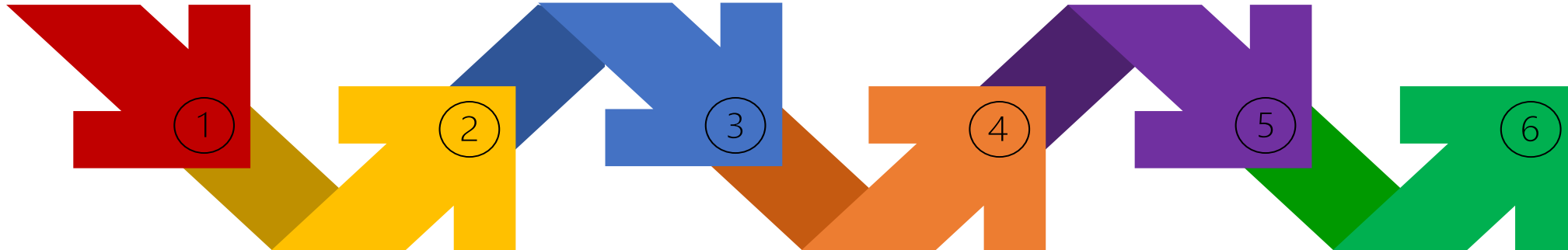
Collect resumes from public datasets or sample data.

Feature Extraction

- Use NLP techniques to extract key information like skills, qualifications, job titles, etc.
- Apply TF-IDF or word embeddings for feature representation.

User Interface

Create a user-friendly UI using **Streamlit** or **Flask** where users can upload resumes and get instant feedback.



Preprocessing

- Use libraries like pdfminer or docx to extract text from resumes.
- Perform text cleaning (removing unnecessary formatting, normalizing text, etc.).

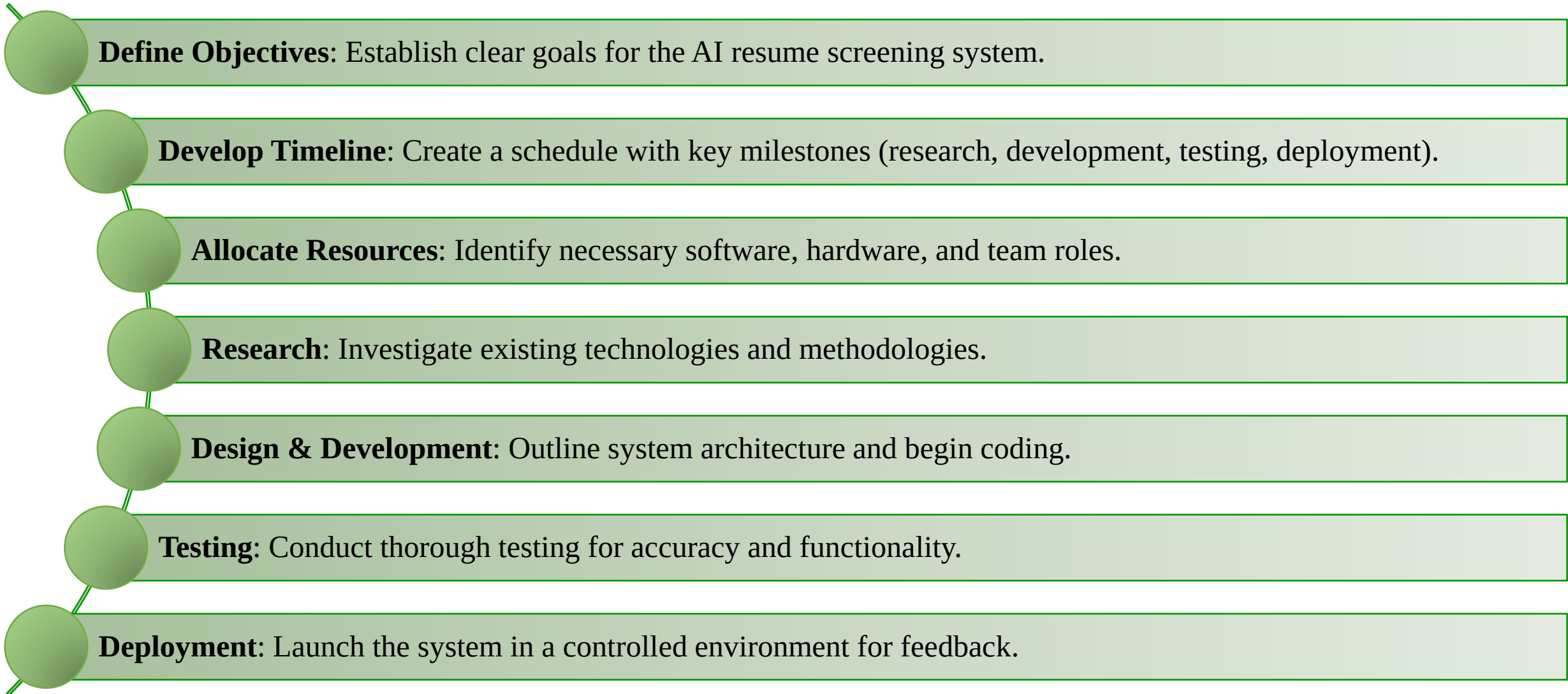
Model Training

Build a machine learning model (e.g., Logistic Regression, SVM, or BERT) to classify resumes based on the extracted features.

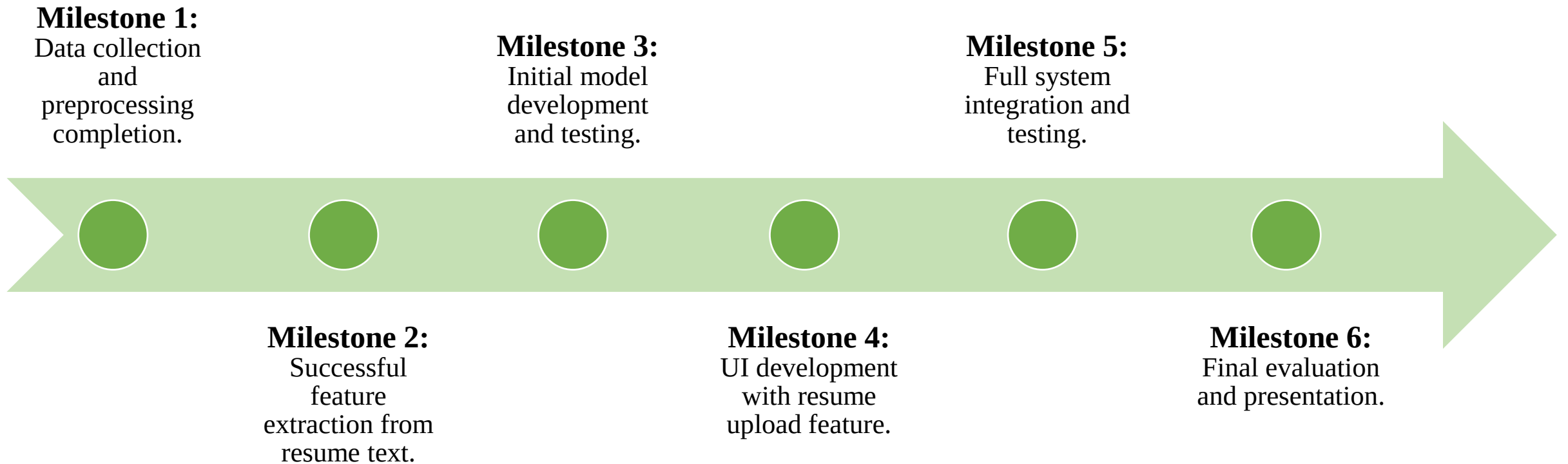
Database

Use SQLite or MongoDB to store resumes and the corresponding analysis.

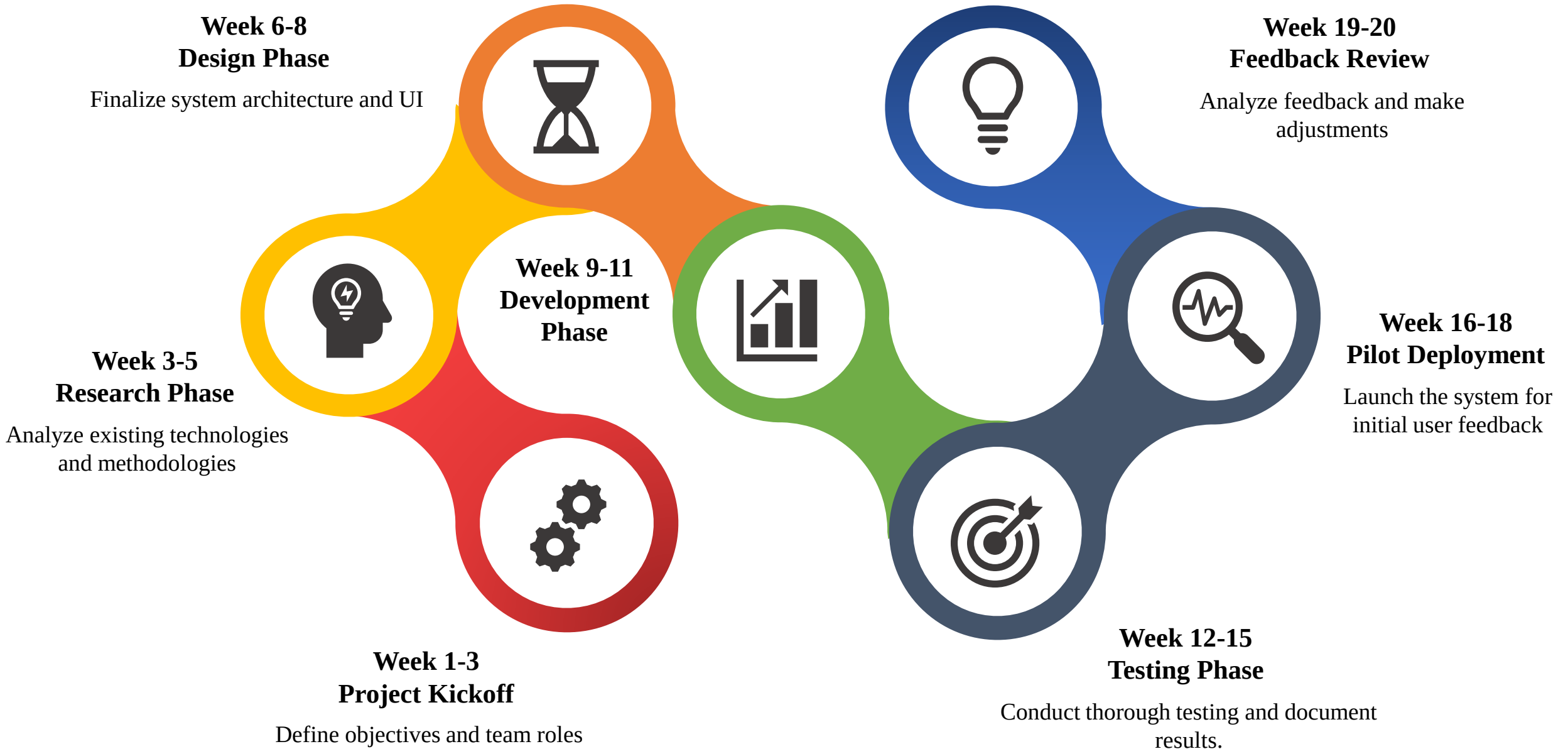
Planning of the Project



Milestones of the Project



Timeline



Expected Outcome

An AI-based system that automates the resume screening process with high accuracy and minimal manual intervention.

A frontend interface that allows recruiters to upload resumes and receive categorized results (shortlisted, rejected, etc.).

A backend model that efficiently extracts and processes unstructured resume data for accurate classification.

Reduced bias and increased efficiency in hiring processes.

Team Members



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